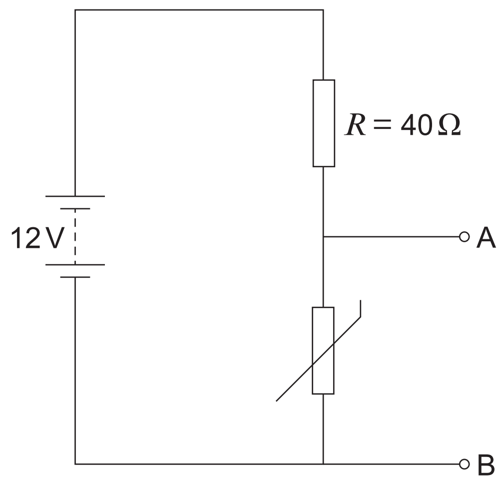
Examiner only

7. The circuit below is used to investigate how the resistance of a thermistor changes as temperature increases.



(a) **Add an ammeter and a voltmeter** to the diagram so the necessary measurements can be taken. [2]

(b) **Plot the results shown** below on the grid opposite and draw a curve of best fit. [2]

Temperature (°C)	Resistance of thermistor (Ω)
0	120
25	92
50	70
75	52
100	40
125	30

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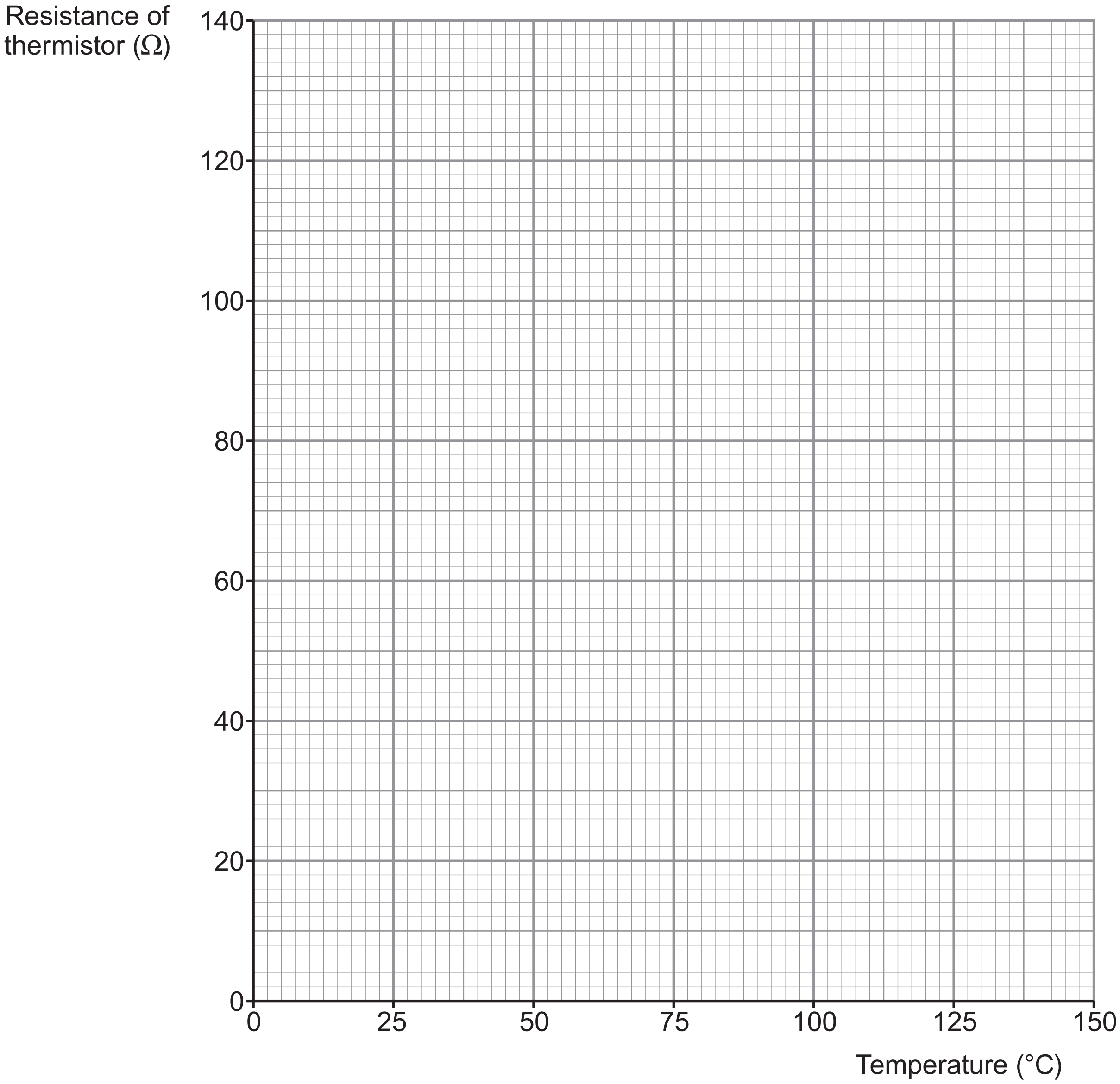


(c) The resistance of the resistor, R, remains constant at $40\,\Omega$.

(i) Calculate the voltage across the terminals AB when the temperature is 100 °C. [3]

voltage = ... V

Examiner only

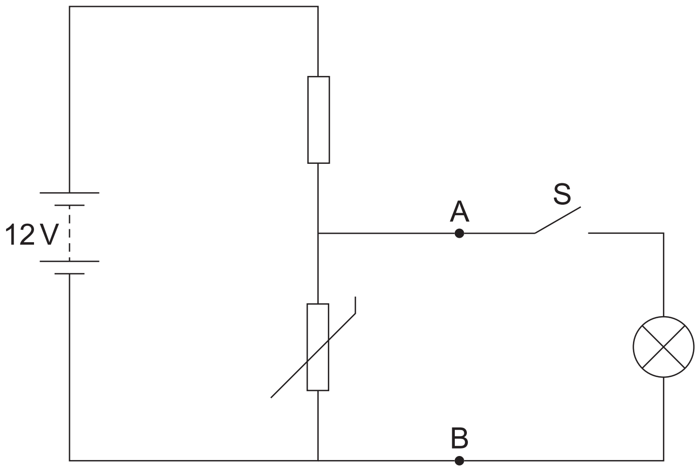
(ii) Tom says as temperature increases the current increases.
Explain whether you agree. [2]

....

....

....

(d) A lamp and a switch are now connected across the terminals AB, as shown in the diagram below.



(i) The lamp has a power of 3 W at 12 V.
Use equations from page 2 to calculate the resistance of the lamp at this power and voltage. [3]

resistance = ... Ω

Examiner only

(ii) The switch, S, is now closed. Explain, without calculation, what happens to the total resistance of the circuit. [2]

....

....

....

....

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Turn over.